Project title: EduGenie-Google Gemini Powered Learning Assistant

Phase 6: Proejct Testing Phase

1. Project Testing:

-Testing is an important phase of the EduGenie project because it ensures that the application works correctly and produces the expected results.

-Testing was performed on different components of the application. The backend, frontend, API communication, Gemini integration, and learning modules were checked individually and as a complete system.

2. Types of Testing Performed

Application Startup Testing:

-The application was started using the FastAPI/Uvicorn server and checked whether the server started successfully.

Frontend Testing:

-The user interface was checked to ensure that the required fields, buttons, and output areas were working correctly.

-Question and Answer Testing:

Different educational questions were entered and the generated responses were checked.

AI Response Testing:

-The communication between the backend and Gemini AI was tested to verify that valid responses were received.

-Quiz Testing:

Different topics were provided as input and the generated quiz content was verified.

-Summary Testing:

Learning content was provided as input and the generated summary was checked for relevance.

-Error Handling Testing:

Invalid or incomplete inputs were tested to ensure that the application handled errors properly instead of stopping unexpectedly.

-API Testing:

The API endpoints were tested to verify that requests and responses were being transferred correctly.

-Through these testing activities, errors and issues were identified and corrected. The final application was verified to ensure that the major functions worked properly.